FCIS – Ain Shams University
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1st Question — Marks: 14

State whether each of the following statements is True of False.

1. While applying down-sampling of an image in the hierarchical LK algorithm, the process of image sharpening would help avoid aliasing.
2. The SIFT feature descriptor is the tool used for extracting image components that are useful for the representation and description of the object shape in an image.
3. The brightness constancy constraint that is utilized in optical flow computation relates the flow to both the spatial and temporal gradients of the image sequence.
4. Normalizing the SIFT descriptors in the direction of the dominant gradient orientation would allow the descriptors to be invariant to image translation.
5. The process of instance segmentation is the process which assigns a different label to the pixels of each object instance in an image.
6. In Content-Based Image Retrieval (CBIR), the retrieval upon features extracted from the former layers of CNN would generate high mean Average Precision (mAP) result for the task of semantic similarity image retrieval.
7. A good action recognition CNN-based model is the one that relies on extracting temporal features only between video frames to define what kind of motion/action occurs.
8. Visual Words help in encoding images by representing them as distribution/histogram of feature occurrences.
9. In the U-Net architecture for semantic segmentation, the up-sampling operation is based mainly on deconvolution operation.
10. In an object detection model where it is supposed to produce a bounding box for each object that appears in the image, you consider a detection output as a false negative if IoU < 0.5.
11. In Harris detector, the decomposition of the eigenvalues λ_1 and λ_2 of the second moment matrix *M* reveals the existence of a corner when $\lambda_1$and λ_2 are large.
12. Suppose two bounding boxes with sizes 2x1 and 2x4 are overlapping in a region of size 1x1. The IoU between these two boxes will equal 1/9.
13. the Hierarchical LK optical flow algorithm tries to solve the problem when the brightness constancy constraint is no longer satisfied.
14. The homography/transformation matrix *H* describes the transformation between points in two related images from different perspective.

2nd Question — Marks: 15

A) [8 Marks] Consider 12 points are painted on a dark plane. 5 points are colored pure red, 5 are colored pure green, and 2 are colored white. Two cameras (stationed with a translational shift) take pictures of the plane. One camera is equipped with a red filter (doesn't see green color) and one camera is equipped with a green filter (doesn't see red color).

- **i.** ***Which*** points does each camera see?
- **ii.** Do you think that enough point correspondences are available to recover a transformation (homography) between the two images? ***Explain*** why.
- **iii.** ***Explain*** an algorithm that can compute the set of good point correspondences between the two images. Define the used parameters of the algorithm, and how the value of each parameter should be selected.

B) [7 Marks] ***Mean Shift*** is a popular method used in computer vision for problems such as image segmentation and object tracking.

- **i.** ***Describe*** the main steps of how Mean Shift is used to perform image segmentation, including the parameter(s) that must be set in order to use Mean Shift.
- **ii.** ***Comment*** on the sensitivity of the algorithm to its parameter value(s); i.e., what is the effect of setting the parameter(s) to values that are too big or too small.
- **iii.** ***Explain*** the advantages of using the Mean Shift algorithm over K-means clustering for image segmentation.

3rd Question — Marks: 20

A) Answer Briefly the following questions:

1. **[2 Marks]** ***What*** is a Gaussian pyramid? Name two applications of it.
2. **[1 Mark]** ***What*** is the direction in the image along which optical flow cannot be reliably estimated?
3. **[2 Marks]** In order to solve the aperture problem of optical flow estimation, the LK algorithm assumes in any given image that nearby pixels move together. It usually constructs a window around each image pixel to construct a system of linear equations in a matrix form which satisfy the brightness constancy equation. ***What*** is the minimum size of a window around each pixel that allows LK to solve the problem? ***Explain*** why.
4. **[2 Marks]** Among the family of region-based object detectors, ***what*** makes the R-CNN very slow for the task of object detections?
5. **[2 Marks]** Mention **One** example of a region-based CNN model that succeeded to enhance the speed of object detections of R-CNN. ***Explain*** how.
6. **[2 Marks]** ***Explain*** the function of Region Proposal Network (RPN) as part of the Faster R-CNN object detector.
7. **[2 Marks]** ***What*** is the typical output size of YOLO as a single-shot detector? ***Show*** what each part of the output indicates.
8. **[1 Mark]** ***Explain*** a modified format of the YOLO output size that allows it to detect multiple classes in each grid cell.

B) [6 Marks] Assume that you have been given images of the road taken from a camera mounted on the front of a car. Your task is to detect and segment cars using instance segmentation. Design a network based on CNN to implement this task. ***Sketch*** and ***explain*** a general architecture diagram in terms of input, core and output components.

4th Question — Marks: 16

A) [8 Marks] Suppose you would like to build a car detection model to use at your garage. You have a database D = {x_1,...,x_N} of the N cars allowed to enter your garage. A training example for your network is a triplet of images (x_1, x_2, x_3) where x_1 and x_2 are different images of the same car, and x_1 and x_3 are images of different cars. The network is supposed to verify any car by computing encodings f_i and f_j to any given two images x_i and x_j where pictures of the same car should have similar encodings.

- **i.** ***Sketch*** a general architecture of this network showing how it would determine whether a car in front of your garage should be allowed to enter.
- **ii.** Briefly ***explain*** the training procedure in terms of input, the learning objective and the loss function used.

B) [8 Marks] Suppose you have a large collection of photos from your trips, including photos of yourself with other people, photos of other people without you, as well as photos without people. Suppose you also have a template of your face, which consists of a small image of your frontal face. ***Describe*** a fully automatic algorithm that uses what you have learned in class to solve this "template matching" problem by finding the subset of the images that contains un-occluded frontal faces of you irrespective of their location, orientation and scale. (Hint: you don't have many example images of yourself except the template image).

With My Best Wishes

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